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MILITARY SPECIFICATION

HEATERS, SPACE, ELECTRIC: BLOWER, RADIANT AND TURRET

(NAVAL SHIPBOARD)

1. SCOPE

1.1 Scope.- This specification covers electric space heaters; blower, radiant and turret, for Naval shipboard use.

1.2 Classification.- Electric heaters shall be furnished in the following types and ratings, as specified (see 6.2). The letters "NM" following the type designation (for example type D-NM) denote heaters of special nonmagnetic construction.

Type B - Blower (2 heat increments-wattage shown applies to "high" heat) Ratings:

2000 watts, 250 (175-355) volts, d.c.
4000 watts, 250 (175-355) volts, d.c.
2000 watts, 500 (355-710) volts, d.c.
4000 watts, 500 (355-710) volts, d.c.
2000 watts, 440 volts, 3 phase, 60 cycles, a.c.
4000 watts, 440 volts, 3 phase, 60 cycles, a.c.

Type D - Radiant (single heat)
Class 1 - Bulkhead mounted
Ratings:

1000 watts, 115 volts, d.c.
1000 watts, 230 volts, d.c.
1000 watts, 440 volts, 3 phase, 60 cycles, a.c.

Class 2 - Portable
Rating:

1000 watts, 115 volts d.c. or single phase, 60 cycle a.c.

Type T - Turret (single heat)
Ratings:

500 watts, 115 volts d.c. or single phase, 60 cycle a.c.
1000 watts, 115 volts d.c. or single phase, 60 cycle a.c.
500 watts, 440 volts, single phase, 60 cycle a.c.
1000 watts, 440 volts, single phase, 60 cycle a.c.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use)
MIL-D-963 - Drawings, Production, Procedure for Procurement of
MIL-B-3743 - Brushes, Electrical Contact; and Carbon Stock, Electrical Contact Brush
MIL-H-22577 - Heating Elements, Electrical; Cartridge, Strip and Tubular Type
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4520

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION, INC.
Standard C33.1-1957 - Flexible Cord and Fixture Wire, Safety Standard for

Standard C73a-1953 - Grounding-Type Attachment Plug Caps and Receptacles and 3-Wire 30-ampere 250 volt Cap and Receptacle

(Application for copies should be addressed to the American Standards Association, Inc., 70 East 45th St., New York 17, N. Y.)

3. REQUIREMENTS

3.1 Material.-

3.1.1 General requirements.- All parts used in the construction of space heaters shall be the lightest possible material consistent with good design and ruggedness of the assembled unit. The heaters shall have no flimsy or closely proportioned parts, and they shall be able to operate satisfactorily for a long life under normal shipboard conditions.

3.1.2 Material not specified hereinafter shall be selected on the basis of meeting the performance requirements of the specification at minimum cost.

3.1.3 Aluminum.- Aluminum, where used, shall be a type alloy that resists corrosion from salt water.

3.1.4 Dissimilar metals.- Aluminum alloys shall not be placed in direct contact with copper or copper bearing alloys.

3.1.5 Nonmagnetic materials.- Heaters of special nonmagnetic construction (see 1.2) shall be constructed entirely of nonmagnetic material unless other material is specifically approved by the bureau or agency concerned. Steel, when used, shall be of a nonmagnetic type. Nonmagnetic material is defined as material which has a magnetic permeability of less than 2.0 after fabrication.

3.1.6 Plastic material.- Plastic material shall be as required by Specification MIL-E-917.

3.1.7 Insulation.- Heat insulating materials shall be highly resistant to heat and moisture absorption. The materials shall be nonfragile and shall be suitably supported.

3.2 Air circulation.- Transfer of heated air from the heater to the ship's space being heated shall be accomplished in an efficient manner. Baffles, deflectors and reflectors may be used to aid in the transfer of heat.

3.3 Load balance.- Heaters for operation on a 440 volt, 3 phase, 60 cycle, a.c. power source shall have three or a multiple of three heating elements of equal wattage connected to provide a balanced load on each phase.

3.4 Connection box.- Heaters shall be provided with a connection box or its equivalent, totally enclosed and separated from the heating element section. The box shall not project from the heater outline. Access to the box shall be obtained by removing only the box cover and not a large section of the heater enclosing case. The connection box shall provide a means for connection of external leads, either a terminal board or a convenient arrangement of heating element terminals so that a direct connection can be made. The external leads shall not enter the connection box through the cover. Except for portable heaters (type D, class 2) that are furnished with a cord and plug, the connection box shall contain a hole or knockout plugs of a size to accomodate the cable clamp furnished with the heater.

3.5 Cable clamps.- A cable clamp shall be provided for securing the cord or supply cable of all heaters, both portable and permanently connected. The clamp shall be capable of securing cable having an outside diameter of 1/4 inch to 5/8 inch. Where knockout plugs are provided, the clamp shall be furnished separately, attached with cord or wire to the heater.

3.6 Connections (within heater).- Connections within the heater shall be made with metal connecting strips or wire suitable for the surrounding temperature. Where the terminal temperature is 400°F or less, wire shall be silicone rubber insulated, tinned copper conductor having an outer covering of glass braid. Where the terminal temperature exceeds 400°F, wire shall be nickel conductor with heavy asbestos insulation. All connections shall be made with nuts and lockwashers.

3.7 Creepage and clearance distances.- The electrical creepage and clearance distances shall be in accordance with Specification MIL-E-917.

3.8 Type B - blower heaters.-

3.8.1 Ratings.- Heaters shall be furnished in the ratings specified in 1.2.

3.8.1.1 Heaters rated at 250 volts, d.c. (see 1.2) shall be capable of satisfactory operation over

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a voltage range of 175 to 355 volts, d.c. At the 355 volt condition, the heater shall be suitable for operating continuously for 4 hours without damage. Heaters with this voltage rating are intended for submarine use. The specified wattage rating applies with a 250 volt, d.c supply voltage.

3.8.1.2 Heaters rated at 500 volts, d.c. shall be capable of satisfactory operation over a voltage range of 355 to 710 volts, d.c. At the 710 volt condition, the heater shall be suitable for operating continuously for 4 hours without damage. Heaters with this voltage rating are intended for submarine use. The specified wattage rating applies with a 500 volt, d.c. supply voltage.

3.8.2 Heat increments.- Heaters shall have 2 heat increments. On d.c. heaters, the low heat shall be approximately one-fourth the wattage of the high heat. High and low heats shall be obtained by parallel, series or series-parallel connection of heating elements in the circuit. On three phase heaters, the low heat shall be approximately one-third the wattage of high heat. High and low heats shall be obtained by delta and wye connection of heating elements in the circuit.

3.8.3 Air temperature rise.- The heater shall be designed so that when operating at rated voltage on high heat, the difference in temperature between air entering and leaving the heater shall be not less than 35°F nor more than 45°F.

3.8.4 Dimensions.- Heater dimensions shall not exceed the following:

Depth - 14-1/2 inches (maximum)
Height - 16-1/2 inches (maximum)
Width - 12-3/8 inches (maximum)

3.8.5 Weights.- Weight shall not exceed 35 pounds for the 2000 watt heaters and 45 pounds for the 4000 watt heater.

3.8.6 Enclosure.- The heaters shall be provided with suitable protective enclosures. The motor and fan shall be located within the heater enclosure for protection against injury. All contactors, relays, and similar items, shall be a part of the assembly and shall be included either in the main enclosure or within a separate enclosure attached to that containing the main assembly.

3.8.7 Grill.- A grill shall be provided on the front of the heater for protection of the elements.

3.8.8 Regulating switch.- Heaters shall be equipped with a 2-heat regulating switch which shall be marked to indicate "high", "low" and "off". The switch shall provide the switching arrangement for proper circuit connection of the heating elements to give the specified heat increments. The switch shall also control the fan motor; the motor being on when the switch is in the "high" and "low" positions and off when the switch is in the "off" position. When the switch is in the "off" position, all lines shall be disconnected. The switch shall be capable of repeatedly interrupting the circuit when switched to any position at maximum voltage, without damage to contacts or insulation.

3.8.9 Heating elements.- Heating elements shall be of the tubular type conforming to specification MIL-H-22577. The heating elements may be provided with fins or cast into finned aluminum grids to improve performance.

3.8.10 Fan.- Air shall be forced from the heater without objectionable noise or vibration, by means of a motor driven fan.

3.8.11 Fan motor.- Motors shall be readily accessible for inspection and repair. The motors shall be suitable for continuous operation at rated voltage. The motors of d.c. heaters rated at 250 and 500 volts (nominal) shall be suitable for 4 hours operation at the maximum voltages of 355 and 710 volts, respectively. None of the motor parts shall be injured from the heat of operations.

3.8.11.1 D.C. motor characteristics.-

3.8.11.1.1 Brushes.- Brushes shall be rectangular in shape and shall conform to Specification MIL-B-3743, except that the brand marking may be omitted. The sides of the brushes shall be smooth and free from saw marks.

3.8.11.1.2 Brush-holders.- Brush-holders shall be readily accessible for the renewal of brush or spring. The brush-feeding device shall insure proper brush contact under all conditions of brush and commutator wear. The brush-holder caps shall be so constructed that no live parts are exposed.

3.8.11.1.3 Commutator.- The commutator shall be of sufficient depth to allow turning down to an extent of 3/32 inch reduction in diameter. Pasted mica shall be used between commutator segments. Pasted mica or suitable molded insulation shall be used between the segments and the shaft or shell. The insulation between commutator bars shall be undercut. The undercutting shall be accomplished by removing all insulation between the bars to form a groove 3/64 inch deep.

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3.8.11.1.4 Frame.- The frame of the motor shall be such that the commutator can be readily inspected, conditioned, and cleaned. This shall be accomplished by means of an easily removable plate over the commutator or by vent holes in the motor frame.

3.8.11.1.5 Connection.- The motor shall be so connected that it will operate at all times when one or more heating elements are energized, and shall be disconnected when the heating elements are disconnected.

3.8.12 Thermal cut-out.- Thermal cut-outs shall be installed which shall operate to disconnect the heater from the power supply if, for any reason, the heater temperature exceeds the safe operating temperature.

3.8.13 Magnetic blow-out.- A magnetic blow-out device shall be used where necessary, to quench the arc when the thermal cut-out operates. The blow-out device shall be sufficiently rugged in design to preclude possibility of damage under any condition of service operation.

3.8.14 Operating temperatures.- The maximum operating temperature shall be reached within 30 minutes starting cold. The temperature rises after 1 hour of continuous operation at rated voltage with the fan running shall not exceed the following:

	Degrees F
Heating element	450
Linoleum under heater	20
Outer surface of case	125

3.8.15 Mounting lugs.- Heaters shall have mounting lugs which provide a space not less than 2 inches between the heater case and the surface upon which it is mounted. Heaters shall be furnished for deck or bulkhead mounting as specified (see 6.2) but provision shall be made on the back or bottom of the heater, as applicable, to mount the heater on either deck or bulkhead by means of reinforcing the bolt holes. The mounting lugs shall support the heater firmly without distortion of the heater base, frame or case.

3.9 Type D - radiant heaters.- Heaters shall consist of an aluminum panel having heating elements cast-in or otherwise enclosed and securely attached, together with the wire guard, frame, connection box and other parts hereinafter described. Bulkhead mounted (class 1) heaters shall have mounting brackets and portable (class 2) heaters shall have legs, carrying handle and cord with plug.

3.9.1 Ratings.- Heaters shall be furnished in the ratings specified in 1.2.

3.9.2 Heat increments.- Heaters shall have one increment of heat.

3.9.3 Design.- The heater shall be designed so as to provide heat by radiation from the front surface of the heating panel and by convection over the front and rear surfaces of the heating panel. A reflecting surface shall be provided behind the heating panel to reflect heat from the rear of the panel. An air space shall be provided between the panel and the reflector.

3.9.4 Heating panel.- The heating panel shall be aluminum alloy with integral fins or a corrugated shape to aid in heat dissipation. Heating elements shall be cast-in or otherwise enclosed and securely attached to the panel.

3.9.5 Heating elements.- Heating elements shall be the tubular type conforming to Specification MIL-H-22577.

3.9.6 Frame.- The heating panel shall be supported by and secured to a metal frame or framework.

3.9.7 Guard.- A substantial corrosion-resisting steel wire guard shall be provided in front of the heating panel to prevent accidental contact with the heated surface.

3.9.8 Connection box.- A connection box shall be located below the heating panel.

3.9.9 Weight.- Total weight of the heater shall not exceed 25 pounds.

3.9.10 Class 1 (bulkhead mounted) heaters.-

3.9.10.1 Dimensions.- The maximum overall dimensions shall be 18 inches by 18 inches by 4 inches deep.

3.9.10.2 Mounting brackets.- Mounting brackets shall be securely attached to the rear of the heater. Four 13/32 inch diameter mounting holes for 3/8 inch bolts shall be provided. The distance between the center lines of the holes shall be 15-5/8 inches horizontally and 16-15/16 inches vertically.

3.9.10.3 Operating temperatures.- The temperature rise after one hour of operation at rated voltage shall not exceed the following:

	Degrees F
Heating panel	400
Bulkhead (steel) directly behind heater	40
Surface of case (including guard)	70

3.9.11 Class 2 (portable) heaters.-

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3.9.11.1 Dimensions.- The maximum overall dimensions shall be 19 inches high by 18 inches wide by 9 inches deep (front to back).

3.9.11.2 Legs and handle.- The heater shall be furnished with a carrying handle and substantial legs to support the heater in an upright position.

3.9.11.3 Switch.- An "ON-OFF" toggle switch shall be mounted on the front of the connection box. The switch shall be a reliable type that mounts in a 1/2 inch hole in the connection box. "ON" and "OFF" positions of the switch shall be clearly marked.

3.9.11.4 Cable assembly and grounding.- The heater shall be provided with a cable and plug connector assembly. The cable shall type SJO, SJT or HSJ conforming with American Standards Association Standard C33.1-1957 and shall extend from the heater 10 feet plus or minus 6 inches. The plug connector shall be a parallel blade, 2 pole, 3 wire, grounding type attachment-plug cap conforming to American Standards Association Standard C73a-1953 and shall be molded to the cable. The grounding conductor of the cable shall be attached to all exposed dead metal parts of the heater at one end and shall be attached at the other end to the grounding blade wiring terminal of the attachment-plug cap.

3.9.11.5 Operating temperature.- The temperature rise at the end of one hour of operation at rated voltage shall not exceed the following:

	Degrees F
Heating panel	400
Deck directly under heater	10
Handle of heater	35
Surface of case (including guard)	70

3.10 Type T - turret heaters.-

3.10.1 Ratings.- Heaters shall be furnished in the ratings specified in 1.2.

3.10.2 Heat increments.- Heaters shall have one increment of heat.

3.10.3 Dimensions.- Heater dimensions shall not exceed the following:

Length - 22 inches (maximum)
Height - 7 inches (maximum)
Width - 6 inches (maximum)

3.10.4 Weights.- Weight shall not exceed 12 pounds for the 500 watt heater and 15 pounds for the 1000 watt heater.

3.10.5 Enclosure.- The sides of the heater case shall be of heavy perforated sheet metal. The ends of the enclosure shall be of solid sheet metal. The enclosure shall be such that current carrying parts cannot be readily short circuited by thrusting objects through the sides of the case. It is recommended but not required that enclosures for both 500 and 1000 watt heaters be identical and that the 500 watt rating be obtained by using half the number of heating elements used in the 1000 watt heater.

3.10.6 Heating elements.- Heating elements shall be of the tubular type conforming to Specification MIL-H-22577. The elements shall be provided with radiating fins.

3.10.7 Operating temperatures.- After 15 minutes of heater operation starting at room temperature, the heating element shall attain at least 80 percent of the temperature rise it attains after 1 hour of operation. The temperature rise after 1-hour operation at rated voltage shall not exceed the following when the heater is mounted on the deck or either horizontally or vertically on a bulkhead.

	Degrees F
Heating element	720
Linoleum under heater	18
Outer surface of case	125

3.10.8 Mounting lugs.- Heaters shall have mounting lugs which provide a space not less than 2 inches between the heater case and the surface upon which it is mounted. Heaters shall be suitable for either deck or bulkhead mounting. The mounting lugs shall support the heater firmly without distortion of the enclosure.

3.10.9 Mounting dimensions.- The heaters shall have four 13/32 inch clearance holes for 3/8 inch bolts in the mounting lugs. The longitudinal dimension between bolt hole centers shall be 20-7/8 inches; the traverse dimension shall be 3 inches between bolt hole centers.

3.11 Painting.- Heaters shall be painted in accordance with Specification MIL-E-917. Heating elements and heating panels or grids shall be left unpainted unless special heat-resisting paint is used to aid in heat transfer and is specifically approved by the bureau or agency concerned. Metals classified as corrosion-resisting materials by Specification MIL-E-917 need not be painted.

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3.12 Identification plate.- Each heater shall be provided with a corrosion-resisting metal identification plate with lettering that is etched, engraved, stamped, cast or photoprocessed. The lettering shall be plainly visible, preferably black in color against a light background. The identification plate shall be securely fastened in a conspicuous location. The plate shall contain the following information:

a. Name of manufacturer
b. Type
c. Voltage
d. Amperes
e. Watts
f. Government stock number (may be left blank if number is not available)
g. Component identification number (may be left blank if number is not available)

3.12.1 Where the heater contains more than one heating element, a wiring diagram showing the method of connection of the elements shall be furnished, either on the identification plate or on a separate plate. The plates shall be located in a position where they may be readily viewed without danger to personnel.

3.13 Insulation resistance.- The insulation resistance of each heater when at normal operating temperature shall be not less than 10 megohms. The insulation resistance when the heater is at normal room temperature shall be not less than 20 megohms.

3.14 Dielectric strength.- All heaters shall be constructed to withstand for a period of one minute, a dielectric test voltage of twice rated voltage plus 1000 volts. The type B, d.c. heaters shall be constructed to withstand for a period of one minute, a dielectric test voltage of twice maximum voltage plus 1000 volts.

3.15 Repair Parts.- No repair parts shall be furnished for type B heaters. For type D and type T heaters, only heating elements in the quantities listed below shall be furnished as repair parts:

No. of identical heating elements installed per ship	No. of heating elements furnished as repair parts
1-4	1
5-8	2
9-20	3
21-50	4
51-100	5
over 100	7

3.16 Drawings.- Drawings shall be in accordance with Specification MIL-D-963. Certification data are not required. Drawings shall contain the following minimum data:

(a) Top, front and side views of the heater assembly with parts identified with piece numbers in the List of Materials. The use of partial cross-section or cutaway views to show method of construction is encouraged.
(b) Mounting dimensions, overall dimensions and location of wiring entrance.
(c) Schematic electrical wiring diagram of heater
(d) Details of terminal connections
(e) Identification plate detail

3.17 Workmanship.- The workmanship shall be first class in every respect. All parts shall be tightly fitted so that when installed where there is vibration, there shall be no rattle of loose parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for acceptance inspection.-

4.2.1 Inspection Lot.- All heaters of the same type, class, voltage and wattage offered for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 Sampling for group A inspection.- A random sample of heaters shall be selected from each inspection lot in accordance with table I and shall be subjected to the group A inspection specified in 4.3.1 with lot acceptance based on the acceptance and rejection numbers of table I.

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Table I - Sampling for group A inspection
AQL (approx.) = 1.5 percent defective.

Number of heaters in inspection lot	Number of heaters in sample	Acceptance number (Number of heaters nonconforming on any group A inspection)	Rejection number (Number of heaters nonconforming on any group A inspection)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

4.2.3 Sampling for group B tests.- A random sample of heaters shall be selected from each inspection lot in accordance with table II and shall be subjected to each of the group B tests specified in 4.3.2.

Table II - Sampling for group B tests.

Number of heaters in inspection lot	Number of heaters in sample	Acceptance number (defectives)	Rejection number (defectives)
Under 16	2	0	1
16 to 40	3	0	1
41 to 65	5	0	1
66 to 110	7	0	1
111 to 200	10	0	1
Over 200	15	0	1

4.2.4 Sampling for group C tests.- The first heater of a particular design, type, class, voltage and wattage shall be subjected to the Group C tests specified in 4.3.3. On subsequent contracts or orders where identical heaters are furnished, the tests need not be repeated, provided that a copy of the original test report, is made available.

4.3 Acceptance inspection.-

4.3.1 Group A inspection.- Each of the sample equipments selected in accordance with 4.2.2 shall be subjected to the examination and tests specified herein, and the results of each test compared with specification requirements. Failure to conform to this specification for any group A inspection shall be counted as a defect and the equipment shall be rejected. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Group A inspection	Reference
Visual and dimensional examination.	4.4.1
Dielectric strength	4.4.2
Insulation resistance (cold)	4.4.3.1

4.3.2 Group B tests.- Each of the sample equipments selected in accordance with 4.2.3 shall be subjected to each of the tests specified herein, and the results of each test compared with specification requirements. Failure to conform to this specification for any group B tests shall be counted as a defect, and the equipment shall be rejected. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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Group B tests	Reference
Operation	4.4.4
Wattage	4.4.5
Insulation resistance(hot)	4.4.3.2

4.3.3 Group C tests.- Each of the sample heaters selected in accordance with 4.2.4 shall be subjected to each of the tests specified herein, and the results of each test compared with the specification requirements. No heater shall be accepted for delivery until these tests have been satisfactorily completed. Failure of the heater to pass any of the tests shall result in rejection of all units so manufactured until the cause of failure has been corrected on all heaters and the sample has been successfully retested. Heaters of the same construction need not be retested on subsequent contracts or orders; therefore, a record of the tests, shall be retained by the manufacturer.

Group C tests	Reference
Weight	4.4.6
Creepage and clearance	4.4.7
Operating temperature	4.4.8
Reliability	4.4.9
Maximum voltage operation	4.4.10 (Type B, d.c. heaters only)

4.4 Examination and test procedures.-

4.4.1 Visual and dimensional examination.- Each of the sample heaters shall be subjected to a thorough examination to ascertain that the materials, design, dimensions and workmanship conform to this specification, contract, and approved drawing.

4.4.2 Dielectric strength test.-

4.4.2.1 General requirements.- The test voltage shall be equal to twice the rated voltage or maximum voltage, as applicable, plus 1,000 volts; the frequency shall be not less than 60 cycles. The voltage wave shall approximate a sine wave. The testing voltage shall be applied continuously for a period not less than 60 seconds.

4.4.2.2 Points of application.- The test voltage shall be successively applied between each electric circuit and the frame, with all other electric circuits and metal parts grounded.

4.4.3 Insulation resistance.- The insulation resistance test shall be made with all circuits of equal voltage above ground connected together. Tests shall be made using a potential of 500 volts, d.c. The insulation resistance shall meet the requirements of 3.13.

4.4.3.1 The insulation resistance (cold) test shall be made at normal room temperature.

4.4.3.2 The insulation resistance (hot) test shall be made immediately following the wattage test.

4.4.4 Operation.- Heaters shall be operated for a minimum of 30 minutes at rated or nominal voltage to determine that their operation is satisfactory. During the test, switches, cut-outs, motors and other components shall be subjected to at least 5 operations each.

4.4.5 Wattage.- The watts input to the heater shall be measured at the end of 15 minutes of continuous operation at nominal or rated voltage and shall be within plus or minus 5 percent of the specified rating. The wattage of type B heaters shall be measured at both heat increments.

4.4.6 Weight.- The heater shall be weighed to determine conformance with 3.8.5 for type B heaters, 3.9.9 for type D heaters or 3.10.4 for type T heaters.

4.4.7 Creepage and clearance.- Creepage and clearance distances shall be demonstrated by actual measurements to be in accordance with Specification MIL-E-917.

4.4.8 Operating temperature.- To determine temperature rise for heaters intended for deck mounting, the heater shall be placed upon a heavy sheet of linoleum during the test. The linoleum shall be not less than 3/32 inch thick and shall extend not less than 1 foot beyond the base of the heater in all directions. For bulkhead mounted heaters, the heater shall be mounted on an upright piece of sheet steel not less than 1/16 inch thick and extending not less than 1 foot beyond the heater. The temperature of the case shall be measured at the hottest part or parts by one or more thermocouples securely fastened to the case. The thermocouples and lead wire shall be not larger than 0.0201 inch (24 Band S gage).

4.4.8.1 Type B heaters.- Heaters shall be tested in the deck mounted position, 2 inches from a wall or bulkhead behind the heater. The heater shall be operated continuously for at least 1 hour at nominal or rated voltage. At the end of 30 minutes, the temperature of the heating element shall be measured and at the end of one hour the temperatures of the heating element, linoleum and case shall be measured to determine compliance with 3.8.14.

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4.4.8.1.1 The temperature of the air leaving the heater shall be measured at nominal or rated voltage with the heater operating at high heat to ascertain compliance with the requirements of 3.8.3.

4.4.8.2 Type D heaters.- Temperature measurements shall be made on Type D heaters at the end of 1 hour of continuous operation at rated voltage to determine compliance with the requirements of 3.9.10.3 for Type D, class 1 heaters or 3.9.11.5 for Type D, class 2 heaters. The class 1 heaters shall be tested as bulkhead mounted heaters and the class 2 heaters shall be placed on linoleum for the test.

4.4.8.3 Type T heaters.- Temperature measurements shall be made on type T heaters after 15 minutes and after one hour of continuous operation at rated voltage to determine compliance with the requirements of 3.10.7. The test shall be conducted with the heater in each of the following positions:

(a) Bulkhead, vertical
(b) Bulkhead, horizontal
(c) Deck

4.4.9 Reliability.- Heaters shall be operated for a total of 100 hours (operating time), under normal conditions, during which time the heater shall be switched "ON" and "OFF" at least 100 times. Each "ON" cycle shall be for a minimum duration of 30 minutes and each "OFF" cycle shall be for a minimum duration of 15 minutes. Type B heaters shall be operated on high heat at rated voltage.

4.4.10 Maximum voltage operation (type B d.c. heaters only).- Type B heaters rated at 250 volts, d.c. (nominal) or 500 volts, d.c. (nominal) (see 1.2) shall be operated at their maximum voltage of 355 or 710 volts, d.c., as applicable, continuously for a period of 4 hours. Following the test, the heater shall be examined for damage to ascertain compliance with 3.8.1.1 and 3.8.1.2.

4.5 Preservation, packaging, and packing.- Sample heaters and packages shall be selected and inspected in accordance with Specification MIL-E-17555 to verify conformance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts.

5.1.1 Heaters.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during handling and direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Shipboard repair parts.- Heating elements furnished as shipboard repair parts shall be individually preserved and packaged level A, packed level C, and marked levels A and C respectively, in accordance with the requirements of Specification MIL-E-22577.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.2).

(5.2.1 The following provides the various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.1.1 Preservation and packaging.-

5.2.1.1.1 Heaters.-

5.2.1.1.1.1 Level A.- Heaters shall be individually preserved and packaged in accordance with method III of Specification MIL-P-116. Switch knobs, handles and other projections shall be adequately cushioned to prevent damage from handling, shipping and storage.

5.2.1.1.1.2 Level C.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.1.2 Repair parts.- Heating elements furnished as repair parts shall be preserved and

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packaged level A in accordance with Specification MIL-H-22577. Heating elements shall be individually packaged.

5.2.1.2 Packing.-

5.2.1.2.1 Heaters.-

5.2.1.2.1.1 Level A.- Heaters shall be packed for overseas shipment in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class 2
PPP-B-640	Class 2

Box closure, strapping or banding shall be in accordance with the applicable box specification or appendix thereto.

5.2.1.2.1.2 Level B.- Each heater shall be packed for domestic shipment and storage in containers conforming to one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Class 1
PPP-B-640	Class 1

Box closure shall be in accordance with the applicable box specification or appendix thereto.

5.2.1.2.1.3 Anchoring, blocking, bracing and cushioning of equipment shall be provided and shall be in accordance with Specification JAN-P-100.

5.2.1.2.2 Repair parts.- Repair parts shall be packed level A or B, as specified (see 6.2) in accordance with Specification MIL-E-17555.

5.2.1.3 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked for shipment in accordance with Standard MIL-STD-129.)

6. NOTES

6.1 Intended use.-

6.1.1 Type B.- Type B, blower heaters, because of lightweight and small space required per kilowatt for efficient compartment heating, are particularly adapted for submarine installations. Heaters should be installed so that the intake air supply is not restricted.

6.1.2 Types D and T.- Types D and T heaters are both intended for use on surface vessels where electrical heating is required. The type D heater is preferred because of the low surface temperature of the heating panel. The type T heater is suitable where space limitations will not permit use of the type D heater.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and rating required (see 1.2).
(c) Class required (type D heaters only) (see 1.2).
(d) Deck or bulkhead mounting required (type B heaters only) (see 3.8.15).
(e) Whether preservation, packaging, packing and marking other than specified in 5.1 are required (see 5.2).
(f) Levels of preservation, packaging and packing and marking required (see 5.2).
(g) Level of packing for repair parts (see 5.2.1.2.2).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs not responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any maner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy-Bureau of Ships
Project 4520-N011Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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